Fortuitous Q-cohomology and quantum lifting: master report

2026-10-26 red = not found in publications

Contents

0. Conventions 1

1. $\mathcal{N}=4$, $SU(N)$ 2

2. $\mathcal{N}=4$, $SO(2N+1)$, $Sp(N)$ 5

3. $\mathcal{N}=2$ $\mathbb{Z}_2$ quiver 7

4. $\mathcal{N}=2$ SCQCD 8

5. Massive-adjoint flows 9

6. Running 11

0. Conventions

Spectral sequence

$$\mathbf{Q}=\sum_{n\geq 0}\hbar^n Q_n,\qquad \sum_{j=0}^{n} Q_j Q_{n-j}=0,\qquad E_1=H(Q_0)$$

$$\lambda=(\theta_1,\theta_2,\theta_3;n_+,n_-),\qquad \Theta=\theta_1+\theta_2+\theta_3,\qquad Y=\#\text{letters},\qquad \mathsf{d}=(1,1,1;-1,-1)$$

$$e_I=\text{unit vector of }\theta_I,\qquad e_\pm=\text{unit vector of }n_\pm$$

$$Q_n:\ C^Y(\lambda)\to C^{Y+1-n}(\lambda-n\mathsf{d}),\qquad d_r:\ E_r^Y(\lambda)\to E_r^{Y+1-r}(\lambda-r\mathsf{d})$$

$$\text{outgoing } d_r:\ r\leq \min_I \theta_I,\qquad \text{incoming } d_r:\ r\leq \min(n_+,n_-)$$

$$d_2[\mathcal{O}_0]=[Q_2\mathcal{O}_0+Q_1\mathcal{O}_1],\qquad Q_0\mathcal{O}_1=-Q_1\mathcal{O}_0$$

$$\mathcal{O}(\hbar)=\sum_{r\geq 0}\hbar^r\mathcal{O}_r,\qquad Q_0\mathcal{O}_r=-R_r,\qquad R_r=\sum_{k=1}^{r}Q_k\mathcal{O}_{r-k}$$

$$\operatorname{rk} d_R=\big(\operatorname{rk} M_R-\operatorname{rk} M_{R-1}\big)-\big(\operatorname{rk} M'_R-\operatorname{rk} M'_{R-1}\big)$$

$$M_R=\Big[\ \underbrace{Q_k\mathcal{K}}_{\text{cocycles}}\ \Big|\ Q_{k-m}:C_m\to C'_k\Big]_{1\leq m\leq k\leq R},\qquad M'_R=M_R\setminus\{Q_k\mathcal{K}\}$$

$$C_m=C^{Y-m}(\lambda-m\mathsf{d}),\qquad C'_k=C^{Y-k+1}(\lambda-k\mathsf{d})$$

Q_n on matrix letters

$$Q_nF=\mathrm{CENG}[n]\,\mu^{-n}\sum_{(i_1,\dots,i_{n+1})}\varepsilon\,A_nK_n\,F\big(\dots,\underbrace{[P,t_{R_1}]}_{i_1},\dots,\underbrace{[t^{R_1},t_{R_2}]}_{i_2},\dots,\underbrace{[t^{R_n},Q]}_{i_{n+1}},\dots\big)$$

$$\mu=-2,\qquad \mathrm{CENG}=(1,1,2,4,8),\qquad t^R=\kappa^{RS}t_S,\qquad \kappa_{RS}=\operatorname{tr}_F(t_Rt_S)$$

Charges

$$E=\tfrac12(\Theta+Y)+n_++n_-,\qquad R_I=\theta_I-\tfrac12(\Theta-Y),\qquad J_L=\tfrac12(\Theta-Y+n_++n_-),\qquad J_R=\tfrac12(n_+-n_-)$$

$$E=R_1+R_2+R_3+2J_L,\qquad 2(E+J_L)=2\Theta+3(n_++n_-)$$

$$(E,R_I,J_L,J_R)(\lambda+k\mathsf{d},\,Y+k)=(E,R_I,J_L,J_R)(\lambda,Y),\qquad (E;R_I;J_L,J_R)(\mathbf{Q})=(\tfrac12;\tfrac12,\tfrac12,\tfrac12;-\tfrac12,0)$$

twisted letter	physical letter	θ	jets	E	(R_1,R_2,R_3)	J_L	J_R
$(\gamma^I)_A$	$\bar\phi_I{}^A$	e_I	–	1	e_I	0	0
$(\beta_I)^A$	$-\frac{i}{\sqrt2}\psi_+^{IA}$	$(1,1,1)-e_I$	–	$\frac32$	$(\frac12,\frac12,\frac12)-e_I$	$\frac12$	0
$\delta^{AB}\mathsf{b}_B$	$\frac14 f_{++}^A$	$(1,1,1)$	–	2	$(0,0,0)$	1	0
$\partial_{+\dot a}\mathsf{c}^A$	$2\bar\lambda_{\dot a}^A$	0	$n_\pm=1$	$\frac32$	$(\frac12,\frac12,\frac12)$	0	$\pm\frac12$
$\partial_{+\dot a}$	$\nabla_{+\dot a}$	0	$n_\pm=1$	1	$(0,0,0)$	$\frac12$	$\pm\frac12$

Quantities

$$h = \dim H^Y(Q_0) \text{ on highest-weight cochains of } (\lambda, Y), \qquad g = \#\text{graviton classes}$$
$$\rho = \mathrm{rk}\big(\mathrm{res}_{\mathfrak{h}} : H^Y \to \text{Cartan restriction}\big)$$
$$0 \to R_N \to C_\infty \xrightarrow{\pi_N} C_N \to 0, \qquad f = h - \mathrm{rk}\,\pi_N^*$$
$$K(Y) = \dim\ker\big(H^Y(R_N) \to H^Y(C_\infty)\big) = f(Y-1)$$

Certification [$\mathbb{Q}$] exact over $\mathbb{Q}$; [p] ranks mod $p = 2^{31}-1$; [p,p′] also $p' = 2147483629$; [cond] $\mathrm{res}(R_r) = 0$ at targets with $h = \rho$.
All orders $\mathbf{Q}\,\mathcal{O}(\hbar) = 0$ (closed completion) $\wedge$ $\mathcal{O}_0 \notin \mathrm{im}\,d_r\ \forall r$.

1. $\mathcal{N} = 4$ Yang–Mills, $G = SU(N)$

1.1 Action and components

$$C^A(\theta) = ig\,\mathsf{c}^A + \theta_I \delta^{AB}(\gamma^I)_B - \tfrac{1}{2}\varepsilon^{IJK}\theta_I\theta_J(\beta_K)^A + \frac{\theta_1\theta_2\theta_3}{ig}\delta^{AB}\mathsf{b}_B, \qquad Q_+^I C^A = \frac{\partial_L C^A}{\partial\theta_I}$$
$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2}\Omega\wedge\Big[\int d^3\theta\Big(\tfrac{1}{2}\delta_{AB}C^A\bar\partial C^B + \tfrac{i}{3!}C_{ABC}C^AC^BC^C\Big)\Big]_{(0,2)}, \qquad Q_0 C^A = \tfrac{i}{2}C^A{}_{BC}C^BC^C$$

$$S_{\mathrm{tw},E} = \int_{\mathbb{C}^2}\Omega\wedge\Big[\mathsf{b}_A\big(\bar\partial\mathsf{c}^A - \tfrac{g}{2}C^A{}_{BC}\mathsf{c}^B\mathsf{c}^C\big) + \big(\bar\partial\gamma_i - ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\big)\beta^i + \bar W(\gamma,\mathsf{w})\Big]_{(0,2)}$$
$$\gamma_i \leftrightarrow (\gamma^I)_A, \qquad \beta^i \leftrightarrow (\beta_I)^A, \qquad (t_A^{\mathrm{adj}})^B{}_C = iC^B{}_{AC}$$
$$\bar W(\gamma,0) = \tfrac{i}{3!}\varepsilon_{IJK}C_{ABC}\delta^{AD}\delta^{BE}\delta^{CF}(\gamma^I)_D(\gamma^J)_E(\gamma^K)_F$$
$$\mathcal{I} = -\tfrac{g}{2}C^A{}_{BC}\mathsf{b}_A\mathsf{c}^B\mathsf{c}^C - ig\mathsf{c}^A(t_A)^j{}_i\gamma_j\beta^i + \bar W(\gamma,\mathsf{w})$$

components: $\underbrace{\mathsf{c}^A,\ \mathsf{b}_A}_{\text{vector}}$, $\underbrace{(\gamma^I)_A,\ (\beta_I)^A}_{I=1,2,3\ \text{adjoint chirals}}$, $A = 1,\dots,N^2-1$; letters $X^{(r,s)} = \partial_{+\dot1}^r\partial_{+\dot2}^s X$, c only with $r+s \geq 1$

1.2 Lie algebra realization

$SU(2)$	basis $(e,h,f) = (E_{12},\ E_{11}-E_{22},\ E_{21})$	$[e,h] = -2e,\ \ [e,f] = h,\ \ [h,f] = -2f$
$SU(3)$	$(e_1,e_2,e_3,h_1,h_2,f_1,f_2,f_3)$	$= (E_{12},E_{23},E_{13},E_{11}-E_{22},E_{22}-E_{33},E_{21},E_{32},E_{31})$
	letter $A = \sum_X A^X t_X,\ \ [t_Y,t_Z] = C^X{}_{YZ}t_X$	$D_Z A^X = C^X{}_{ZY}A^Y,\ \ \kappa_{XY} = \mathrm{tr}(t_X t_Y)$
	singlets	$\mathrm{wt} = 0\ \cap\ \ker D_{f_1}\cap\ker D_{f_2}\quad (SU(2)\text{: } \mathrm{wt} = 0\cap\ker D_f)$
	charge conjugation	$Ct_X = -t_X^{\mathsf{T}},\ \ C = \pm$ split $\frac{1}{2}(1\pm C)$
$SU(N)$	fundamental $N\times N$, tr $= 0$	$\{E_{ij}\}_{i\neq j}\cup\{E_{ii}-E_{i+1,i+1}\},\ \ t^R = \kappa^{RS}t_S$
	Casimir tensor	$\sum_R (t_R)_{ab}(t^R)_{cd} = \delta_{ad}\delta_{bc} - \frac{1}{N}\delta_{ab}\delta_{cd},\ \ C_F = \frac{N^2-1}{N}$
	probes	$x \in \mathfrak{sl}_N(\mathbb{F}_p)$: N^2-1 random entries, $x_{NN} = -\sum_{i<N} x_{ii}$; $\mathrm{Tr}\,\mathbf{1} = N$
	trace words ($N \geq 4$)	$C_\infty = \mathbb{C}[\mathrm{tr}(w)]_{\lvert w\rvert\geq 2},\ \ R_N = \ker\big(\pi_N : C_\infty \to C_N\big)$ (trace identities)

1.3 Highest-weight algorithm

(a) $SU(2)$, $SU(3)$: bosonic highest weights

$$G_{\mathrm{HW}} = GL(3)_\theta \times SL(2)_{\dot a}\ \ \text{(letter-wise)}, \qquad [G_{\mathrm{HW}}, Q_n] = 0$$
$$\theta(\gamma^I) = e_I, \quad \theta(\beta_I) = (1,1,1) - e_I, \quad \theta(\mathsf{b}) = (1,1,1), \quad \theta(\mathsf{c}) = 0; \qquad (n_+,n_-)\big(X^{(r,s)}\big) = (r,s)$$
$$\lambda \text{ dominant}: \ \theta_1 \geq \theta_2 \geq \theta_3 \geq 0, \qquad n_+ \geq n_- \geq 0$$
$$C^Y_{\mathrm{HW}}(\lambda) = \big\{\text{colour singlets, weight } \lambda,\ Y \text{ letters}\big\}\cap\ker E_{12}\cap\ker E_{23}\cap\ker J_+, \qquad J_+ = u\,\partial_v \ \text{ on } \ X^{(r,s)} \leftrightarrow u^r v^s$$
$$C^Y_{\mathrm{HW}}(\lambda) = \bigoplus_{\text{skeletons}} C^Y_{\mathrm{HW}}(\lambda;S), \qquad S = \{(\text{type}, r+s)\}\ \text{multiset}$$
$$\dim C^Y_{\mathrm{HW}}(\lambda;S) = \text{Weyl-alternated colour-singlet character (exact)}$$

$$SU(3): \quad g = \mathrm{Mult}\big(t_F\otimes t_B\otimes\tau\big), \qquad t_F:\ \gamma\mapsto V,\ \beta_I\mapsto e_{I+1}\wedge e_{I+2},\ \mathsf{b}\mapsto e_1\wedge e_2\wedge e_3,\ \mathsf{c}\mapsto 1\ \ \text{(column wedges of } \theta)$$
$$t_B = \prod_{n_-}[pq]\prod u, \quad [pq] = u_pv_q - v_pu_q, \qquad \tau = \text{colour trace word}, \qquad (g\pm Cg)(x_k)\ \text{at probes}\ \Rightarrow\ V^\pm$$

$$h = \dim C^Y_{\mathrm{HW}} - \operatorname{rk} Q_0|_{Y-1} - \operatorname{rk} Q_0|_Y, \qquad \rho: \text{ evaluation at diagonal probes}$$

(b) $SU(N \geq 4)$: $\mathfrak{gl}(3|2)$ super highest weights

$$V = \mathbb{C}^{3|2} = \langle \theta_1, \theta_2, \theta_3 \,|\, z_1, z_2 \rangle, \qquad \text{letters of degree } k: \ \Pi^{k-1}\Lambda^k V, \qquad L = \Theta + n_+ + n_-$$

$$\mu \vdash L, \quad \mu_4 \leq 2 \ \ ((3|2) \text{ hook})$$

$$\lambda(\mu) = \big(\mu_1, \mu_2, \mu_3; \ \#\{i \geq 4 : \mu_i \geq 1\}, \ \#\{i \geq 4 : \mu_i \geq 2\}\big)$$

raising: E_{12}, E_{23}, J_+ (even), $\theta_f \partial_z$ (odd), distinguished Borel

$$K(\mu, Y) = \dim \ker\big(H^Y(R_N)_\mu \to H^Y(C_\infty)_\mu\big)$$

$$K = \dim R^Y - \operatorname{rk} \begin{pmatrix} Q: R^Y \to R^{Y+1} & 0 \\ \iota: R^Y \to C^Y_\infty & Q_0: C^{Y-1}_\infty \to C^Y_\infty \end{pmatrix} + \operatorname{rk} Q_0\big|_{C^{Y-1}_\infty} - \operatorname{rk} Q\big|_{R^{Y-1}}$$

stable range: $R^j_N = 0 \ \ (j < 2N+2)$; routes: $\ell(\mu) \leq 2$, hooks: theorems; else exact Q and stable inclusion

1.4 Frontier

G	all sectors complete	partial / targeted
$SU(2)$	$E + J_L \leq \frac{33}{2}$ [p]	detector all sectors $\leq \frac{35}{2}$; $\frac{37}{2}$–22 partial; chains at $E + J_L = \frac{39}{2}, 20$
$SU(3)$	$E + J_L \leq 15$ [p] ($\leq \frac{23}{2}$: 2610.20340)	detector ≤ 17; windows at 16, $\frac{33}{2}$, 17, $\frac{35}{2}$
$SU(4)$	$E + J_L \leq \frac{27}{2}$ [p]	14: 34/38 shapes (incl. $K = 1$)
$SU(5)$	≤ 14	$\frac{29}{2}$: 26/29; 15: 3/53
$SU(6)$	≤ 15	$\frac{31}{2}$: 1/38
$SU(7)$	≤ 15	$\frac{31}{2}$: 19/37; 16: 14/62
$SU(8)$	$\leq \frac{31}{2}$	16: 45/56; $\frac{33}{2}$: 19/45
$SU(9)$	$\leq \frac{33}{2}$	17: 41/64; $\frac{35}{2}$: 18/50
$SU(10)$	≤ 17	$\frac{35}{2}$: 33/38; 18: 39/72
$SU(11..15)$	≤ 18	–

1.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1, R_2, R_3)	J_L	J_R	data
$SU(2)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	primary; 2209.06728
	$\mathcal{O}_0$ multiplet	$E + J_L \in [\frac{25}{2}, \frac{33}{2}]$	7					69 HW blocks (descendants)
	$\mathcal{O}_1$	$(6,6,6;0,0)$	9	$\frac{27}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	primary; 2304.10155
	products	$E + J_L \in [16, \frac{33}{2}]$	9					graviton $\times \mathcal{O}_0$, descendants: fresh 0
	P39	$(5,5,5;3,0)$	9	15	$(2,2,2)$	$\frac{9}{2}$	$\frac{3}{2}$	$h = 13$, known 12, fresh 1 [p,p′]
	X40	$(7,5,5;1,1)$	10	$\frac{31}{2}$	$(\frac{7}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{9}{2}$	0	$(h, \rho, g) = (8,8,7)$ [p,p′]
$SU(3)$	X_7	$(2,2,2;2,2)$	7	$\frac{21}{2}$	$(\frac{5}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{3}{2}$	0	$C = -1$, $(h, \rho, g) = (26, 26, 25)$; 35 HW blocks
	X_7 member	$(0,0,0;5,5)$	7	$\frac{27}{2}$	$(\frac{7}{2}, \frac{7}{2}, \frac{7}{2})$	$\frac{3}{2}$	0	fortuitous [$\mathbb{Q}$]; 26 trace generators
	F_1 tower	$(n{+}4,4,4;0,0)$	$n{+}7$	$n{+}\frac{19}{2}$	$(n{+}\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	$n \geq 3$; 2412.08695, 2610.20340
	B_1 tower	$(n{+}5,5,4;0,0)$	$n{+}8$	$n{+}11$	$(n{+}2,2,1)$	3	0	2412.08695, 2610.20340
	new primary	$(5,4,4;1,1)$	10	$\frac{27}{2}$	$(\frac{7}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{5}{2}$	0	$C = -1$, $h = 2$, $\rho = 0$ [cond: free]
	new primary	$(7,4,4;1,0)$	11	14	$(5,2,2)$	$\frac{5}{2}$	$\frac{1}{2}$	$C = +1$, $h = 2$ (F_1 desc.+1), $\rho = 0$
	new primary	$(6,4,4;1,1)$	11	$\frac{29}{2}$	$(\frac{9}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{5}{2}$	0	$C = +1$, $h = 3$, $\rho = 0$
	new primary	$(7,5,4;1,0)$	11	$\frac{29}{2}$	$(\frac{9}{2}, \frac{5}{2}, \frac{3}{2})$	3	$\frac{1}{2}$	$C = +1$, $h = 4$, $\rho = 0$
$SU(4)$	first class	$(4,2,2;2,2)$	9	$\frac{25}{2}$	$(\frac{9}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{3}{2}$	0	$C = -1$, $K = 1$ [$\mathbb{Q}$]; first $SU(4)$ class
	(Codex)	$(2,2,2;3,3)$	9	$\frac{27}{2}$	$(\frac{7}{2}, \frac{7}{2}, \frac{7}{2})$	$\frac{3}{2}$	0	17500-term stable primitive; $= (4,2,2;2,2) - 2e_1 + e_+ + e_-$
$SU(5..15)$	–							none through the frontier

descendant shifts at fixed Y: $(-e_I; +e_\pm)$ $(\theta\, \partial_z)$, $(+e_I; 0)$ (∂_θ), $(7,4,4;1,0)Y11 \to (6,4,4;1,1)Y11$: $(-e_1; +e_-)$

1.6 Quantum lift

class	result	level
$\mathcal{O}_0$ ($SU(2)$)	closed completion $\sum_{r \leq 4} \hbar^r \mathcal{O}_r$; no incoming $\Rightarrow$ all orders	[$\mathbb{Q}$]

class	result	level
$\mathcal{O}_1$ $(SU(2))$	$d_1 \neq 0$ (2606.27955)	[p,p′]
P39	$d_1 = 0$; $R_2, R_3, R_4 \in \operatorname{im} Q_0$; $r = 5$ target $h = 0$; no incoming ⇒ all orders	[p,p′]
X40	$d_1 = 0$; $R_2 \in \operatorname{im} Q_0 + Q_1 H$; $R_3, R_4 \in \operatorname{im} Q_0$; $[R_5] = 0$; incoming $h = 0$ ⇒ all orders	[p,p′] ($r \leq 2$), [p]
X_7	$d_1 = 0$ in/out; d_2 out target $h = 0$, in source $h = 0$; $r \geq 3$ empty ⇒ all orders	[p]
F_1, $n = 3$	closed completion $\mathcal{O}_0..\mathcal{O}_4$; $r \geq 5$ empty; no incoming ⇒ all orders ($d_1 = 0$: 2610.20340)	[p]
$(5,4,4;1,1)Y10$	$d_1 = d_3 = 0$, $d_4 = 0$ [cond], no incoming; d_2: target $(3,2,2;3,3)Y9$, $h = 331$, $\rho = 329$	running
$(7,4,4;1,0)Y11$	$d_1 = 0$ (target $h = 0$), $d_2 = 0$ [cond] ($h = \rho = 1464$), $d_3 = 0$, $d_4 = 0$ [cond], no incoming ⇒ all orders	[p], [cond]
$(6,4,4;1,1)Y11$	no incoming; $d_3 = 0$; $d_4 = 0$ [cond]; d_1, d_2 open	[p]
$(7,5,4;1,0)Y11$	no incoming; $d_4 = 0$ [cond] ($h = \rho = 501$); d_1, d_2, d_3 open	[p]
B_1, $(8,5,4;0,0)Y11$	d_1 target $(7,4,3;1,1)Y11$: $h = 2$, $\rho = 0$	open
$SU(4)$, $(4,2,2;2,2)Y9$	d_1: R_1 exact at $(3,1,1;3,3)Y9^-$ ($h = 111$); R_2 exact ($h = 0$); $r \geq 3$ empty; incoming d_1 rank 0 (453 sources), d_2 source $h = 0$ ⇒ all orders	[p]
(Codex) $(2,2,2;3,3)Y9$	five Q_1 scalar queries $= 0$	partial

Chain data: P39, $(5,5,5;3,0)Y9$

r	target	target data	verdict
1	$(4,4,4;4,1)$, 9	$h = 13$	$d_1 = 0$
2	$(3,3,3;5,2)$, 8	$g_8 = 583$	$R_2 \in \operatorname{im} Q_0$ (45,450,714 terms)
3	$(2,2,2;6,3)$, 7	$h = 1$	$R_3 \in \operatorname{im} Q_0$ (17,787,235)
4	$(1,1,1;7,4)$, 6	$h = \rho = g = 152$	$R_4 \in \operatorname{im} Q_0$ (1,276,152)
5	$(0,0,0;8,5)$, 5	$h = 0$	–

r	X40 target	data	verdict
1	$(6,4,4;2,2)$, 10	–	$d_1 = 0$ (4,429,603)
2	$(5,3,3;3,3)$, 9	–	$R_2 \in \operatorname{im} Q_0 + Q_1 H$
3	$(4,2,2;4,4)$, 8	$g_8 = 1112$	$R_3 \in \operatorname{im} Q_0$
4	$(3,1,1;5,5)$, 7	–	$R_4 \in \operatorname{im} Q_0$
5	$(2,0,0;6,6)$, 6	$h = \rho = g = 30$	$\operatorname{res} R_5 = 0$

Chain data: F_1, $n = 3$, $(7,4,4;0,0)Y10$, $C = -1$

r	target	$\dim / h / \rho$	residual	$\mathcal{O}_r$ (coordinates; $\operatorname{rk} Q_0$)
1	$(6,3,3;1,1)$, 10	34430/0/0	$Q_1\mathcal{O}_0$	$Y9$ (28311; 18003)
2	$(5,2,2;2,2)$, 9	63983/609/608	$Q_2\mathcal{O}_0 + Q_1\mathcal{O}_1$	$Y8$ (49328; 32144)
3	$(4,1,1;3,3)$, 8	26400/0/0	$Q_2\mathcal{O}_1 + Q_1\mathcal{O}_2$	$Y7$ (18919; 13063)
4	$(3,0,0;4,4)$, 7	1896/34/34	$Q_3\mathcal{O}_1 + Q_2\mathcal{O}_2 + Q_1\mathcal{O}_3$	$Y6$ (1169; 825)

r	$SU(4)$ $(4,2,2;2,2)Y9$: target, $C = -1$	data
1	$(3,1,1;3,3)$, 9	18133/12574/5448, $h = 111$
2	$(2,0,0;4,4)$, 8	983/666/317, $h = 0$
in 1	$(5,3,3;1,1)$, 9	$h = 453$, $\operatorname{rk} d_1 = 0$
in 2	$(6,4,4;0,0)$, 10	$h = 0$

Structural

$$\operatorname{res}(\hbar)\,\mathbf{Q} = \mathbf{Q}_{\mathrm{IR}}\,\operatorname{res}(\hbar), \quad \mathbf{Q}_{\mathrm{IR}} = 0 \;\Rightarrow\; \operatorname{res}(R_r) = 0 \;\Rightarrow\; d_r = 0 \text{ at targets with } h = \rho \quad ([\mathrm{cond}];\ r = 1:\ 2610.22332)$$

$$SU(2),\ Y = 11,\ E + J_L \in [\tfrac{41}{2}, \tfrac{45}{2}]:\ \ 7 \text{ blocks},\ \dim E_2 = 17,\ E_2 \subset \operatorname{span}(E_2 \cdot E_2)$$

$$\rho - g = 1:\ \mathrm{X40},\ X_7 \quad (\text{Coulomb-visible fortuitous})$$

$$0 \neq [X_7] \in H^7\big(\mathfrak{sl}_3[z_+, z_-], \mathfrak{sl}_3; \mathbb{C}\big)_{(5,5)}, \qquad X_7(z) = 4838400 \quad [\mathbb{Q}]$$

1.7 Data

$SU(2)$ per level $\quad \text{states} = \sum h \cdot \dim(\theta)_{SU(3)} \cdot (n_+ - n_- + 1)$

$E{+}J_L$	sect.	HW	B	F	NG blk	NG	$E{+}J_L$	sect.	HW	B	F	NG blk	NG
1	1	0	0	0	0	0	9	39	112	1786	1173	0	0
$\frac{3}{2}$	1	0	0	0	0	0	$\frac{19}{2}$	26	140	1542	2934	0	0
2	2	1	6	0	0	0	10	46	211	3327	3507	0	0
$\frac{5}{2}$	1	1	0	6	0	0	$\frac{21}{2}$	39	271	6618	3734	0	0
3	5	2	1	8	0	0	11	59	413	6807	8733	0	0
$\frac{7}{2}$	2	2	18	0	0	0	$\frac{23}{2}$	46	531	9582	13824	0	0
4	6	4	18	12	0	0	12	81	811	21483	13593	1	1
$\frac{9}{2}$	5	4	2	38	0	0	$\frac{25}{2}$	59	1053	26406	25614	1	1
5	9	6	36	30	0	0	13	94	1563	30504	46380	3	3
$\frac{11}{2}$	6	7	102	18	0	0	$\frac{27}{2}$	81	2028	63444	49640	4	5
6	16	15	59	139	0	0	14	116	2986	90111	74934	7	9
$\frac{13}{2}$	9	17	96	228	0	0	$\frac{29}{2}$	94	3835	101250	138786	7	11
7	19	29	423	114	0	0	15	150	5567	177657	170608	12	19
$\frac{15}{2}$	16	37	402	420	0	0	$\frac{31}{2}$	116	7203	280314	222792	9	19
8	26	60	345	912	0	0	16	172	10330	333117	391821	14	27
$\frac{17}{2}$	19	72	1230	714	0	0	$\frac{33}{2}$	150	13377	494548	548270	15	30

NG = odd-Y (non-graviton) highest-weight classes: all in the $\mathcal{O}_0$ multiplet at $Y=7$; $Y=9$ blocks at $E+J_L=16, \frac{33}{2}$: products/descendants (fresh 0).

$SU(N\geq 4)$ relation scan

shape μ at level $2(E+J_L)=2(\mu_1+\mu_2+\mu_3)+3\sum_{i\geq 4}\mu_i$

K one prime; $\quad K(4,2,2,2,2)=1$ [$\mathbb{Q}$], $\quad$ all other completed shapes: $K=0$

$SU(4)$, $\mu=(4,2,2,2,2)$, Y	5	6	7	8	9	10	11	12
$\dim R^Y$	17	213	1201	3430	5321	4579	2085	404
$\mathrm{rk}\, Q\vert_{R^Y}$	17	196	1005	2424	2894	1681	401	0
$\dim H^Y(R_4)$	0	0	0	1	3	4	3	3
K	0	0	0	0	0	**1**	0	0

$K = 4579 - 7900 + 6216 - 2894 = 1$ [$\mathbb{Q}:$ 359/467/783 primes > Hadamard], $\quad SU(5),\ \mu=(4,2,2,2,2):\ K=0$

2. $\mathcal{N}=4$ Yang–Mills, $G=SO(2N+1),\ Sp(N)$

2.1 Action and components

$S_{\mathrm{tw},E}$ and Q_0 of 1.1, $\quad C_{ABC}$ of $\mathfrak{so}(2N+1)$, $\mathfrak{sp}(N)$, $\quad A=1,\dots,\dim G$; $\quad$ components: $\mathsf{c}^A, \mathsf{b}_A, (\gamma^I)_A, (\beta_I)^A$

2.2 Lie algebra realization

$SO(2n+1):\ N_F=2n+1,\quad A^{\mathsf{T}}\eta+\eta A=0,\quad \eta_{i\bar{i}}=1;\qquad Sp(n):\ N_F=2n,\quad A^{\mathsf{T}}\Omega+\Omega A=0,\quad \Omega_{i\bar{i}}=s_i$

$\bar{\imath}=N_F+1-i,\qquad s_i=+1\ (i\leq n),\ \ s_i=-1\ (i>n),\qquad \mathfrak{h}=$ diagonal

basis: independent subset of $E_{ij}-E_{\bar{\jmath}\bar{\imath}}$ (SO), $\qquad E_{ij}-s_is_jE_{\bar{\jmath}\bar{\imath}}$ (Sp)

$\kappa_{RS}=\mathrm{tr}_F(t_Rt_S),\qquad \sum_R (t_R)_{ab}(t^R)_{cd}=\frac{1}{2}\delta_{ad}\delta_{bc}-\frac{1}{2}F_{ac}F_{bd},\quad F=\eta$ or Ω, $\qquad C_F=\frac{N_F-1}{2}$ (SO), $\frac{N_F+1}{2}$ (Sp)

trace words: $O(\infty)$ and $Sp(\infty)$ stable algebras coincide (reversal identity); $\quad C_\infty$ realized on $SO(n),\ n\geq$ #letters

2.3 Highest-weight algorithm

$G_{\mathrm{HW}},\ \lambda$ dominant, $C^Y_{\mathrm{HW}}(\lambda)$ as 1.3(a)

$\dim C^Y_{\mathrm{HW}}:$ Weyl integration, torus points summed over signed-permutation orbits (exact)

$\dim C^Y_\infty:$ Burnside over $\prod S_2\wr S_k$ on perfect matchings

$$f_Y=h_Y-\mathrm{rk}\,\pi^*,\qquad \mathrm{rk}\,\pi^*=\mathrm{rk}\begin{pmatrix}\Pi & B_G\\ Q_0^\infty & 0\end{pmatrix}-\mathrm{rk}\,Q_0^\infty-\mathrm{rk}\,B_G$$

stable range: $Y<N_F+1$ (SO), $\ Y<n+1$ $(Sp(n))$ $\Rightarrow$ $f=0$

2.4 Frontier

(one prime, all sectors)

G	complete $E+J_L \le$	partial (done/total sectors)
SO(3)	12	13: 16/94, 14: 11/116
Sp(1)	12	13: 16/94, 14: 11/116
SO(5)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(2)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
SO(7)	$\frac{23}{2}$	12: 10/81, 13: 16/94, 14: 11/116
Sp(3)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(9)	$\frac{23}{2}$	12: 17/81, 13: 16/94, 14: 11/116
Sp(4)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116
SO(11)	12	$\frac{25}{2}$: 17/59, 13: 48/94, $\frac{27}{2}$: 1/81, 14: 25/116
Sp(5)	12	$\frac{25}{2}$: 13/59, 13: 32/94, $\frac{27}{2}$: 1/81, 14: 12/116

$SO(3) = Sp(1)$: $SU(2)$ census ($\le \frac{33}{2}$).

2.5 Fortuitous generators and classes

G	class	λ	Y	E	(R_1, R_2, R_3)	J_L	J_R	data
$SO(3) = Sp(1)$	$\mathcal{O}_0$	$(4,4,4;0,0)$	7	$\frac{19}{2}$	$(\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$	$\frac{5}{2}$	0	first level
$SO(5) = Sp(2)$	first class	$(3,2,2;1,1)$	7	9	$(3,2,2)$	1	0	first level; $f=1$, $\rho=0$
$SO(7)$	O_f	$(3,3,3;0,0)$	8	$\frac{17}{2}$	$(\frac{5}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{1}{2}$	0	first level; 2506.13887; $(h,\rho)=(1,1)$
$Sp(3)$	first class	$(5,2,2;1,1)$	9	11	$(5,2,2)$	1	0	first level; $h=f=1$
$Sp(3)$	ω	$(5,0,0;3,3)$	9	13	$(7,2,2)$	1	0	$h=1$, $\rho=0$; $\lambda = (5,2,2;1,1) - 2e_2 - 2e_3 + 2e_+ + 2e_-$
$SO(9)$	first class	$(5,3,3;0,0)$	10	$\frac{21}{2}$	$(\frac{9}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{1}{2}$	0	first level; $h=6=5+1$
$SO(11)$	class	$(7,3,3;0,0)$	12	$\frac{25}{2}$	$(\frac{13}{2}, \frac{5}{2}, \frac{5}{2})$	$\frac{1}{2}$	0	$h=23$, $f=1$ ($E+J_L=\frac{25}{2}$ partial)
$Sp(4), Sp(5)$	–							none through $E+J_L \le 12$

$$SO(2N+1),\ N=3,4,5: \quad \lambda = (2N-3,3,3;0,0),\quad Y = 2N+2,\quad E = \tfrac{4N+5}{2},\quad J_L = \tfrac{1}{2}$$

hits above the first level (2.7): shifts $(-e_I; +e_\pm)$, $(+e_I; 0)$ at fixed Y of the first class

2.6 Quantum lift

class	result	level
O_f ($SO(7)$)	d_1 rank 1 into $O_m = (2,2,2;1,1)Y8$ ($h=2, \rho=1$); 2511.09519	[p] (three primes)
$SO(7)$ $(4,3,3;0,0)Y8$	d_1 rank 1 into $(3,2,2;1,1)Y8$ ($h=11$, $\rho=10$)	[p]
$SO(7)$ $(3,1,1;2,2)Y8$	$d_1 \neq 0$ into $(2,0,0;3,3)Y8$ ($h=2, \rho=1$); $Sp(3)$: $d_1=0$	[$\mathbb{Q}$]
$SO(5)$ $(3,2,2;1,1)Y7$	out d_1, d_2 targets $h=0$; in d_1 source $h=0$; $r \ge 3$ empty $\Rightarrow$ all orders	[p]
$SO(9)$ $(5,3,3;0,0)Y10$	d_1 rank 1 into $(4,2,2;1,1)Y10$ ($h=10$, $\rho=9$)	[p]
$SO(11)$ $(7,3,3;0,0)Y12$	d_1, d_2, d_3 submitted	running
$Sp(3)$ $(5,2,2;1,1)Y9$	out d_1, d_2; in d_1 from $(6,3,3;0,0)Y9$: submitted	running
ω	no outgoing; in d_1, d_3: sources $h=0$; in d_2: from $(7,2,2;1,1)Y10$ ($h=454$)	open
$SO(7)$	$h-\rho=1$; in d_1 from $(7,2,2;1,1)Y10$: rank 0 ($SO(7)$ and $Sp(3)$); in d_2 source	[p]
$\beta = (6,1,1;2,2)Y10$	$h=0$	

after one loop: $(h,\rho)^{SO(2N+1)} = (h,\rho)^{Sp(N)}$ in all four lifted sectors

2.7 Data

h (computed sectors) / **f** per level * stable-range sectors not counted in h; ^{p} partial level

$E+J_L$	$SO(3)$	$Sp(1)$	$SO(5)$	$Sp(2)$	$SO(7)$	$Sp(3)$	$SO(9)$	$Sp(4)$	$SO(11)$	$Sp(5)$
8	60	60	164	164	43*	203	0*	207	0*	207
$\frac{17}{2}$	72	72	207	207	117*	255	0*	261	0*	261
9	112	112	381	381	385 / **1***	495	0*	516	0*	516
$\frac{19}{2}$	140	140	495	495	596 / **1***	647	0*	675	0*	675
10	211	211	877 / **1**	877 / **1**	1218 / **3***	1230	168*	1304	0*	1309
$\frac{21}{2}$	271	271	1151 / **2**	1151 / **2**	1641 / **5***	1636	554*	1736	0*	1745
11	413	413	1996 / **6**	1996 / **6**	3047 / **9**	3029	2058 / **1***	3277	0*	3308
$\frac{23}{2}$	531	531	2665 / **9**	2665 / **9**	4099 / **11**	4077	3627 / **2***	4426	0*	4470
12	811 / **1**	811 / **1**	90 / **1**^{p}	90 / **1**^{p}	100^{p}	7408 / **1**	0^{p}	8252	651*	8374
$\frac{25}{2}$	–	–	–	–	–	1385^{p}	–	1748^{p}	1642^{p}	1821^{p}
13	261^{p}	261^{p}	726 / **7**^{p}	726 / **7**^{p}	868^{p}	1380^{p}	842^{p}	1659^{p}	848 / **1**^{p}	1741^{p}
$\frac{27}{2}$	–	–	–	–	–	24^{p}	–	44^{p}	50^{p}	50^{p}
14	160^{p}	160^{p}	378 / **6**^{p}	378 / **6**^{p}	429^{p}	424^{p}	434^{p}	442^{p}	13^{p}	447^{p}

(h,ρ) of lifted pairs

sector	$SO(2N+1)$	$Sp(N)$
$(3,3,3;0,0)Y8$	$(1,1)$	$(0,0)$
$(2,2,2;1,1)Y8$	$(2,1)$	$(1,1)$
$(4,3,3;0,0)Y8$	$(7,7)$	$(6,6)$
$(3,2,2;1,1)Y8$	$(11,10)$	$(10,10)$
$(5,3,3;0,0)Y10$	$(6,6)$	$(5,5)$
$(4,2,2;1,1)Y10$	$(10,9)$	$(9,9)$

ω orbit	$SO(7)$ h/ρ	$Sp(3)$ h/ρ
$(5,0,0;3,3)Y9$	$0/0$	$1/0$
$(6,1,1;2,2)Y10$	$355/354$	$354/354$
$(7,2,2;1,1)Y11$	$1/0$	$1/0$
$(8,3,3;0,0)Y12$	$41/40$	$41/40$

3. $\mathcal{N}=2$ $\mathbb{Z}_2$ quiver, $G = SU(N) \times SU(N)$

3.1 Action and components

$$\sigma = \mathrm{diag}(\mathbf{1}_N, -\mathbf{1}_N), \qquad S_{\mathrm{tw},E} \text{ of 1.1 for } \mathfrak{u}(2N), \text{ restricted to } \sigma X \sigma^{-1} = X \ (\mathsf{c}, \mathsf{b}, \gamma^3, \beta_3), \quad \sigma X \sigma^{-1} = -X \ (\gamma^{1,2}, \beta_{1,2})$$

$$\mathsf{c}, \mathsf{b}, \gamma^3, \beta_3 = \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \in \mathfrak{su}(N) \oplus \mathfrak{su}(N), \qquad \gamma^{1,2}, \beta_{1,2} = \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \in (\mathbf{N}, \bar{\mathbf{N}}) \oplus (\bar{\mathbf{N}}, \mathbf{N})$$

$$\kappa^2 = g_2^2/g_1^2 : \quad Q_0 \ni \kappa^2 \text{ on the node-2 block of the hypermultiplet moment maps}$$

$$Q_n \text{ weights: } \prod_{\text{inputs}} w \prod_{\text{vertices}} \text{coef} \prod_{\text{lines}} \text{prop}$$

$$\#\kappa^2 \text{ in a term of } Q_n = \#(\mathsf{b}_2, \beta_{3,2} \text{ consumed}) - \#(\mathsf{b}_2, \beta_{3,2} \text{ produced})$$

$$\Rightarrow \ Q_n(\kappa) = \phi_\kappa Q_n(1) \phi_\kappa^{-1} \ \ \forall n \ \Rightarrow \ E_r(\kappa) \cong E_r(1) \ \ (\kappa \neq 0)$$

3.2 Lie algebra realization

$$N_F = 2N, \qquad \mathfrak{g} = \left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : \mathrm{tr}\, a = \mathrm{tr}\, b = 0 \right\}, \qquad \text{bifundamental lines: } E_{ij} \ (i \leq N < j \text{ or } j \leq N < i), \text{ dual } E_{ji}$$

node-2 generators: $t^R \to \kappa^2 t^R$; weights: $U(N) \times U(N)/U(1)_{\mathrm{diag}}$ torus $\Rightarrow$ baryon-number-zero invariants

3.3 Highest-weight algorithm

$$G_{\mathrm{HW}} = GL(2)_{12} \times GL(1)_3 \times SL(2)_{\dot{a}}, \qquad \theta_1 \geq \theta_2 \geq 0, \ \ \theta_3 \geq 0, \ \ n_+ \geq n_- \geq 0$$

$$C^Y_{\mathrm{HW}}(\lambda) = \{\text{colour singlets}\} \cap \ker E_{12} \cap \ker J_+, \qquad \text{columns: } \theta_1 \text{ wedge columns of } (\theta_1, \theta_2) \ \oplus \ \theta_3 \text{ single slots}$$

3.4 Frontier

$SU(2)^2 : \ E + J_L \leq 10$ (all sectors); $\qquad SU(3)^2 : \ E + J_L \leq \frac{17}{2}$; $\qquad$ running: $SU(2)^2 \ E + J_L \in [\frac{21}{2}, 14]$, $SU(3)^2$ $[9, 10]$

3.5 Classes and quantum lift

Classes $\quad f$: open (no stable comparison); E_2 classes per level in 3.6

$$E_2 = h - \mathrm{rk}\, d_1^{\mathrm{out}} - \mathrm{rk}\, d_1^{\mathrm{in}}; \qquad d_2 \text{ candidates: } E_2^Y(\lambda) > 0, \ E_2^{Y-1}(\lambda - 2\mathsf{d}) > 0$$

$$\mathrm{rk}\, d_2 = 0 \ \text{ on every candidate pair, } \ \kappa^2 = 1, 9$$

$$SU(2)^2 : \ 29 \text{ pairs } (Q_2 \mathcal{K} \neq 0 : \ 21), \qquad SU(3)^2 : \ 15 \text{ pairs } (Q_2 \mathcal{K} \neq 0 : \ 3)$$

$$d_3 : \ (3,3,3;0,0)Y6 \to (0,0,0;3,3)Y4 : \ \mathrm{rk} = 0 \ \ (Q_3 \mathcal{K} = 0)$$

$$\text{sources } (E + J_L \leq 9) : \ \rho = 0$$

$$\text{targets } (h, \rho) : \ (1,1,0;2,2)Y4 \ (6,4), \ (2,0,0;2,2)Y4 \ (2,0), \ (1,1,1;2,2)Y4 \ (20,14), \ (2,0,1;2,2)Y4 \ (6,0)$$

$$\text{control: } Q_2 \to 4Q_2 \ \Rightarrow \ \mathrm{rk}\, d_2 = 3$$

3.6 Data

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_2 pairs	$\max \mathrm{rk}\, d_2$	$\#\{Q_2\mathcal{K}\neq 0\}$ $(\kappa^2=1,9)$
2	2	4	3	4	0	4	0	–	–
2	$\frac{5}{2}$	2	1	2	0	2	0	–	–
2	3	8	4	6	0	6	0	–	–
2	$\frac{7}{2}$	4	4	6	0	6	0	–	–
2	4	13	11	17	0	17	0	–	–
2	$\frac{9}{2}$	8	7	14	0	14	0	–	–
2	5	20	19	38	1	36	0	–	–
2	$\frac{11}{2}$	13	15	42	1	40	0	–	–
2	6	31	34	93	2	89	0	–	–
2	$\frac{13}{2}$	20	26	116	1	112	0	–	–
2	7	44	57	240	3	230	1	0	0
2	$\frac{15}{2}$	31	43	276	3	268	1	0	0
2	8	61	90	560	5	544	3	0	4
2	$\frac{17}{2}$	44	73	716	4	696	1	0	2
2	9	84	140	1298	7	1258	7	0	10
2	$\frac{19}{2}$	61	110	1690	8	1640	5	0	8
2	10	111	205	3153	13	3059	11	0	18
3	2	4	3	4	0	4	0	–	–
3	$\frac{5}{2}$	2	1	2	0	2	0	–	–
3	3	8	5	8	0	8	0	–	–
3	$\frac{7}{2}$	4	5	8	0	8	0	–	–
3	4	13	12	21	0	21	0	–	–
3	$\frac{9}{2}$	8	8	18	0	18	0	–	–
3	5	20	24	54	0	54	0	–	–
3	$\frac{11}{2}$	13	19	62	0	62	0	–	–
3	6	31	40	149	1	147	1	0	0
3	$\frac{13}{2}$	20	32	176	1	174	0	–	–
3	7	44	69	400	4	390	2	0	0
3	$\frac{15}{2}$	31	54	478	3	468	1	0	0
3	8	61	113	1016	8	986	8	0	4
3	$\frac{17}{2}$	44	92	1304	7	1270	3	0	2

4. $\mathcal{N}=2$ SCQCD, $SU(N)$, $N_f=2N$

4.1 Action and components

$$\kappa\to 0 \text{ of } 3.1: \quad \underbrace{\mathsf{c},\mathsf{b},\gamma^3,\beta_3}_{\text{node 1 only}}\in\mathfrak{su}(N)$$

$$\gamma^{1,2},\beta_{1,2}=\begin{pmatrix}0&**&0\end{pmatrix}:\ 2N \text{ fundamental hypermultiplets}, \qquad SU(N)_2 \text{ flavour singlets}$$

4.2 Lie algebra realization and highest weights

$$\mathfrak{g}=\left\{\begin{pmatrix}a&0\\0&0\end{pmatrix}:\operatorname{tr}a=0\right\}, \qquad \text{node-2 roots: Haar measure only;} \qquad G_{\mathrm{HW}} \text{ as } 3.3$$

4.3 Frontier

$$SU(2),\ N_f=4:\ E+J_L\le\tfrac{23}{2}; \qquad SU(3),\ N_f=6:\ E+J_L\le 9; \qquad \text{running: } SU(2)\ E+J_L=12$$

4.4 Classes and quantum lift

Classes f: open (no stable comparison); E_2 classes per level in 4.5

$$SU(2):\ d_1=0 \text{ on every sector } E+J_L\le\tfrac{23}{2}$$
$$SU(2):\ \mathrm{rk}\,d_2=0 \text{ on every candidate pair } (28;\ Q_2\mathcal{K}\neq 0:\ 27)$$
$$SU(3):\ d_1\neq 0:\ (1,1,5;0,0)Y6,\ (1,1,4;1,0)Y6,\ (1,1,3;1,1)Y6,\ (1,1,6;0,0)Y6$$
$$(1,1,5;1,0)Y6,\ (1,1,4;1,1)Y6,\ (1,1,4;2,0)Y6,\ (1,1,7;0,0)Y8$$
$$SU(3):\ \mathrm{rk}\,d_2=0 \text{ on every candidate pair } (13;\ Q_2\mathcal{K}\neq 0:\ 5)$$

4.5 Data

N	$E{+}J_L$	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_2 pairs	$\max \mathrm{rk}\, d_2$	$\#\{Q_2\mathcal{K}\neq 0\}$
2	2	4	3	3	0	3	0	–	–
2	$\frac{5}{2}$	2	1	1	0	1	0	–	–
2	3	8	3	4	0	4	0	–	–
2	$\frac{7}{2}$	4	3	4	0	4	0	–	–
2	4	13	7	10	0	10	0	–	–
2	$\frac{9}{2}$	8	4	6	0	6	0	–	–
2	5	20	9	21	0	21	0	–	–
2	$\frac{11}{2}$	13	9	21	0	21	0	–	–
2	6	31	19	37	0	37	0	–	–
2	$\frac{13}{2}$	20	14	54	0	54	0	–	–
2	7	44	26	112	0	112	0	–	–
2	$\frac{15}{2}$	31	23	109	0	109	0	–	–
2	8	61	42	232	0	232	1	0	0
2	$\frac{17}{2}$	44	30	307	0	307	1	0	1
2	9	84	59	481	0	481	1	0	1
2	$\frac{19}{2}$	61	54	623	0	623	1	0	1
2	10	111	87	1263	0	1263	4	0	4
2	$\frac{21}{2}$	84	75	1527	0	1527	5	0	5
2	11	144	122	2632	0	2632	8	0	8
2	$\frac{23}{2}$	111	106	3850	0	3850	7	0	7
3	2	4	3	3	0	3	0	–	–
3	$\frac{5}{2}$	2	1	1	0	1	0	–	–
3	3	8	4	5	0	5	0	–	–
3	$\frac{7}{2}$	4	4	5	0	5	0	–	–
3	4	13	9	12	0	12	0	–	–
3	$\frac{9}{2}$	8	6	8	0	8	0	–	–
3	5	20	14	25	0	25	0	–	–
3	$\frac{11}{2}$	13	12	26	0	26	0	–	–
3	6	31	26	55	0	55	0	–	–
3	$\frac{13}{2}$	20	23	66	0	66	0	–	–
3	7	44	40	150	1	148	1	0	0
3	$\frac{15}{2}$	31	35	161	1	159	1	0	0
3	8	61	69	323	2	319	2	0	0
3	$\frac{17}{2}$	44	56	424	1	420	2	0	1
3	9	84	101	781	3	773	7	0	4

5. Massive-adjoint flows

5.1 Action and components

$$\bar{W}\to \bar{W}+\tfrac{\Lambda}{2}\,\delta^{AB}(\gamma^3)_A(\gamma^3)_B, \qquad Q_0'=Q_0+\Lambda\,\delta^{AB}(\gamma^3)_B\frac{\partial}{\partial(\beta_3)^A}, \qquad Q_n'=Q_n\ (n\geq 1)$$

$$\Lambda = M \text{ (one node)}, \qquad \Lambda=\mathrm{diag}(M_1\mathbf{1}_N, M_2\mathbf{1}_N) \text{ ($\mathbb{Z}_2$ quiver)}$$

UV	mass	IR
$\mathcal{N}=4$ $SU(2)$, $SU(3)$	M on γ^3	Leigh–Strassler; $\bar{W}_{\rm IR}=-\frac{1}{2M}\,\mathrm{Tr}[\gamma^1,\gamma^2]^2$
SCQCD $SU(2)$, $SU(3)$	M on γ^3	$\mathcal{N}=1$ SQCD, $N_f=2N$, quartic $\bar{W}$
$\mathbb{Z}_2$ quiver $SU(2)^2$	$(M_1,M_2)=(M,-M)$	Klebanov–Witten

5.2 Gradings and charges

$$(A,B;n_+,n_-;\mathrm{gh}):\quad A=\theta_1+\theta_3,\quad B=\theta_2+\theta_3,\quad \mathrm{gh}=Y-\Theta; \qquad C(A,B;n_\pm;\mathrm{gh})=\bigoplus_{\theta_3}C^Y(\lambda)$$

$$R_1+R_3=A+\mathrm{gh},\quad R_2+R_3=B+\mathrm{gh},\quad J_L=\tfrac{1}{2}(n_++n_--\mathrm{gh}),\quad J_R=\tfrac{1}{2}(n_+-n_-)$$

$$R_{\rm IR}=\tfrac{1}{2}(R_1+R_2)+R_3, \qquad E_{\rm IR}=\tfrac{3}{2}R_{\rm IR}+2J_L, \qquad w=A+B+2(n_++n_-)=2R_{\rm IR}+4J_L=\tfrac{4}{3}\big(E_{\rm IR}+J_L\big)$$

$$d_r:\ (A,B;n_\pm;\mathrm{gh})\to(A-2r,B-2r;n_\pm+r;\mathrm{gh}+1+2r), \qquad w \text{ fixed}$$

$$G_{\rm HW}=GL(2)_{12}\times SL(2)_{\dot a}:\quad A\geq B\geq 0,\ \ n_+\geq n_-\geq 0; \qquad \text{Lie algebra as 1.2, 3.2, 4.2}$$

5.3 Frontier

theory	complete
$\mathcal{N}=4$ $SU(2)$, one mass	$w \le 16$
$\mathcal{N}=4$ $SU(3)$, one mass	$w \le 12$
SCQCD $SU(2)$ + mass	$w \le 13$
SCQCD $SU(3)$ + mass	$w \le 11$
$\mathbb{Z}_2$ quiver $SU(2)^2$, $(M,-M)$	$w \le 13$

5.4 Classes and quantum lift

theory	class	$(A,B;n_+,n_-;\mathrm{gh})$	$(R_1{+}R_3, R_2{+}R_3)$	J_L	R_{IR}	E_{IR}	quantum
$SU(2)$, one mass	w	$(3,3;0,0;-2)$	$(1,1)$	1	1	$\frac{7}{2}$	graviton
	w^2	$(6,6;0,0;-4)$	$(2,2)$	2	2	7	closed completion to $\hbar^3$ (quartic $\bar{W}$)
	$\mathcal{O}_7$	$(5,2;0,0;-2)$	$(3,0)$	1	$\frac{3}{2}$	$\frac{17}{4}$	$d_1=0$; $r\ge 2$ empty $\Rightarrow$ all orders [p]
	$\mathcal{O}_{16}$	$(8,8;0,0;-5)$	$(3,3)$	$\frac{5}{2}$	3	$\frac{19}{2}$	all orders [p]

$$\mathcal{O}_7, \mathcal{O}_9, \mathcal{O}_{14}, \mathcal{O}_{16}: \ 2512.12674$$

$$\mathcal{O}_{16}: \ d_1 \ (h_T=0), \ d_2 \ (h_T=7, \ Q_2\mathcal{K}\neq 0), \ d_3 \ (h_T=8), \ d_4 \ (Q_4\mathcal{K}=0)=0; \ \text{no } d_5, \ \text{no incoming}$$

$$SU(2), \ w\le 16: \ d_1\neq 0 \text{ only at } (6,6;1,1;-3); \qquad \mathrm{rk}\, d_2 = \mathrm{rk}\, d_3 = 0 \text{ on every candidate pair}$$

$$\mathbb{Z}_2 \text{ quiver}: \ d_1\neq 0 \text{ at } (2,2;k,1;1), \ k=1,2,3$$

$$\text{other theories: } d_1=0, \qquad \mathrm{rk}\, d_2 = \mathrm{rk}\, d_3 = 0$$

Massive SQED analogue

theory	$(7,7;1,0;-5)$	$(7,7;1,0;-4)$	$(7,7;1,0;-6)$
$U(1)$, $\mathcal{I}=ig\mathsf{c}(\gamma_+\beta^+-\gamma_-\beta^-)+\frac{m}{2}\gamma_s^2+y\gamma_s\gamma_+\gamma_-$	$d_2\neq 0$ (closed form)		
$\mathcal{N}=4$ $SU(2)$, one mass	$h=0$	$h=1$, $d_2=0$ $(Q_2\mathcal{K}\neq 0)$	$h=0$
$\mathcal{N}=4$ $SU(3)$, one mass	$h=0$	running	
SCQCD $SU(2)$ + mass	$E_1=0$		

5.5 Data

theory	w	sectors	HW blocks	$\sum h$	$\#\{d_1\neq 0\}$	$\sum E_2$	d_r pairs	r tested
$\mathcal{N}$=4 $SU(2)$, one mass	4	7	4	4	0	4	–	–
	5	7	4	4	0	4	–	–
	6	13	11	11	0	11	–	–
	7	13	12	13	0	13	–	–
	8	22	25	30	0	30	–	–
	9	22	24	40	0	40	–	–
	10	34	51	81	0	81	–	–
	11	34	49	115	0	115	–	–
	12	50	93	218	0	218	4	d_2
	13	50	90	317	0	317	4	d_2
	14	70	154	569	0	569	11	d_2
	15	70	151	845	0	845	14	d_2, d_3
	16	95	240	1473	1	1471	32	d_2, d_3
$\mathcal{N}$=4 $SU(3)$, one mass	4	7	6	6	0	6	–	–
	5	7	10	10	0	10	–	–
	6	13	19	23	0	23	–	–
	7	13	27	42	0	42	–	–
	8	22	50	96	0	96	1	d_2
	9	22	58	171	0	171	1	d_2
	10	34	98	358	0	358	5	d_2
	11	34	112	641	0	641	7	d_2
	12	50	178	1284	0	1284	15	d_2
SCQCD $SU(2)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	16	46	0	46	–	–
	7	13	0	0	0	0	–	–
	8	22	34	194	0	194	–	–
	9	22	0	0	0	0	–	–
	10	34	62	826	0	826	2	d_2
	11	34	0	0	0	0	–	–
	12	50	109	3371	0	3371	7	d_2
	13	50	0	0	0	0	–	–
SCQCD $SU(3)$ + mass	4	7	8	11	0	11	–	–
	5	7	0	0	0	0	–	–
	6	13	18	44	0	44	–	–

theory	w	sectors	HW blocks	$\sum h$	$\#\{d_1 \neq 0\}$	$\sum E_2$	d_r pairs	r tested
	7	13	0	0	0	0	–	–
	8	22	35	203	0	203	1	d_2
	9	22	0	0	0	0	–	–
	10	34	71	934	0	934	3	d_2
	11	34	0	0	0	0	–	–
$\mathbb{Z}_2$ quiver $SU(2)^2$, masses $(M, -M)$	4	7	10	16	0	16	–	–
	5	7	0	0	0	0	–	–
	6	13	26	66	0	66	–	–
	7	13	0	0	0	0	–	–
	8	22	53	299	1	297	–	–
	9	22	0	0	0	0	–	–
	10	34	98	1306	1	1304	3	d_2
	11	34	0	0	0	0	–	–
	12	50	170	5442	1	5440	13	d_2, d_3
	13	50	0	0	0	0	–	–

6. Running

item	computation
$SU(3)$ $(5,4,4;1,1)Y10$	d_2 chain: $Q_1\mathcal{O}_0$, $Q_2\mathcal{O}_0$, solve $\mathcal{O}_1$, $Q_1\mathcal{O}_1$, solve R_2 (generator split)
$Sp(3)$ $(5,2,2;1,1)Y9$; $SO(11)$ $(7,3,3;0,0)Y12$	d_1, d_2, in d_1; d_1, d_2, d_3
$SO(7)$ β	d_1 from $(7,2,2;1,1)Y10$ (recheck), chains
SO/Sp scans	$Sp(3..5)$, $SO(11)$: $E + J_L \in [\frac{25}{2}, 14]$
$SU(4)$, $SU(5)$	last shapes at $E + J_L = 14$, $\frac{29}{2}$
quiver, SCQCD	$SU(2)^2$ to $E + J_L = 14$, $SU(3)^2$ to 10, SCQCD $SU(2)$ 12; d_3 scans
massive flows	w scans (all five theories); $SU(3)$ one mass $(7,7;1,0;-4)$ d_2
$SU(3)$ β/LS (d_{abc})	vertex-resolved Q_n tables, d_2